# FP4 QAT: MXFP4 Experts + Indexer QK

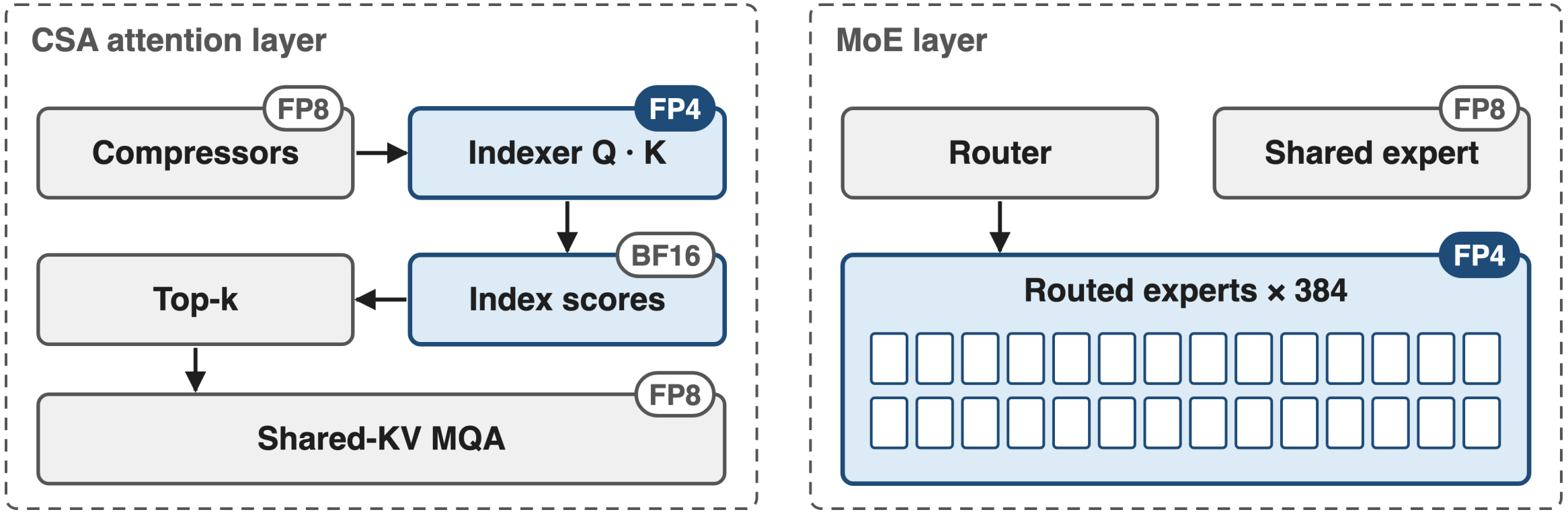
## Train against FP4 after pre-training, serve the same bits

DeepSeek-AI · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Low-Precision & Quantization · arxiv.org/abs/2606.19348

# V4 runs two hot spots in FP4




Applied as QAT in post-training; other linear weights stay FP8 (E4M3, 128 × 128 blocks)

# Expert weights are nearly all of V4

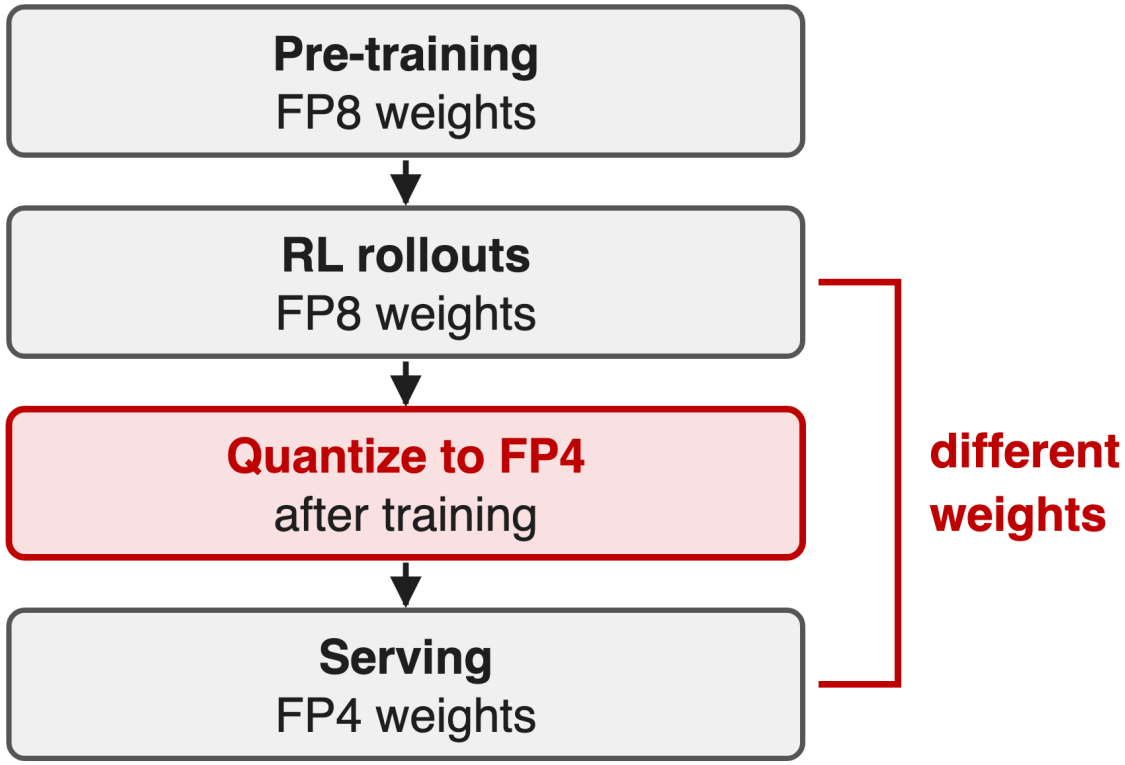
**~97%**

of V4-Pro's 1.6T parameters are routed-expert weights: ~1.55 TB even in FP8

*Analytic: 61 layers × 384 experts × 3 matrices × 7,168 × 3,072 = 1.547T parameters (V4 §4.2.1); 1 byte per FP8 weight*

# FP8 is too big; FP4 after training drifts

- **Stay FP8**: 1 B per expert weight
- **Quantize after RL**: rollouts sample a model nobody serves
- **FP8 indexer, FP32 scores**: the 1M-token scan stays heavy




**Villain: training stacks speak FP8, so FP4 arrives after training, when nothing can adapt**

# Key idea: FP4 → FP8 dequantization is exact

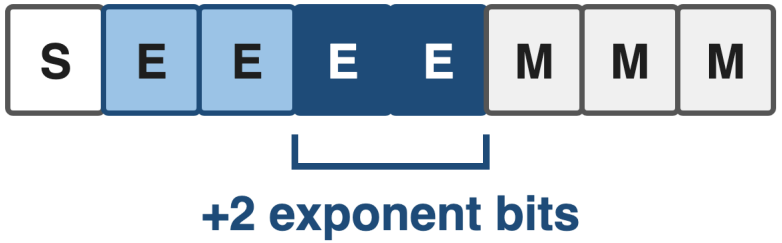
**FP4**

E2M1

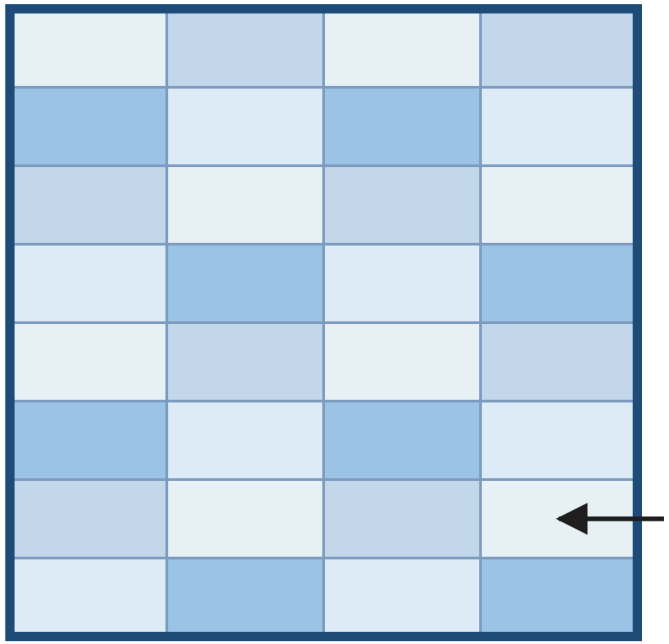



**FP8**

E4M3




S sign · E exponent · M mantissa



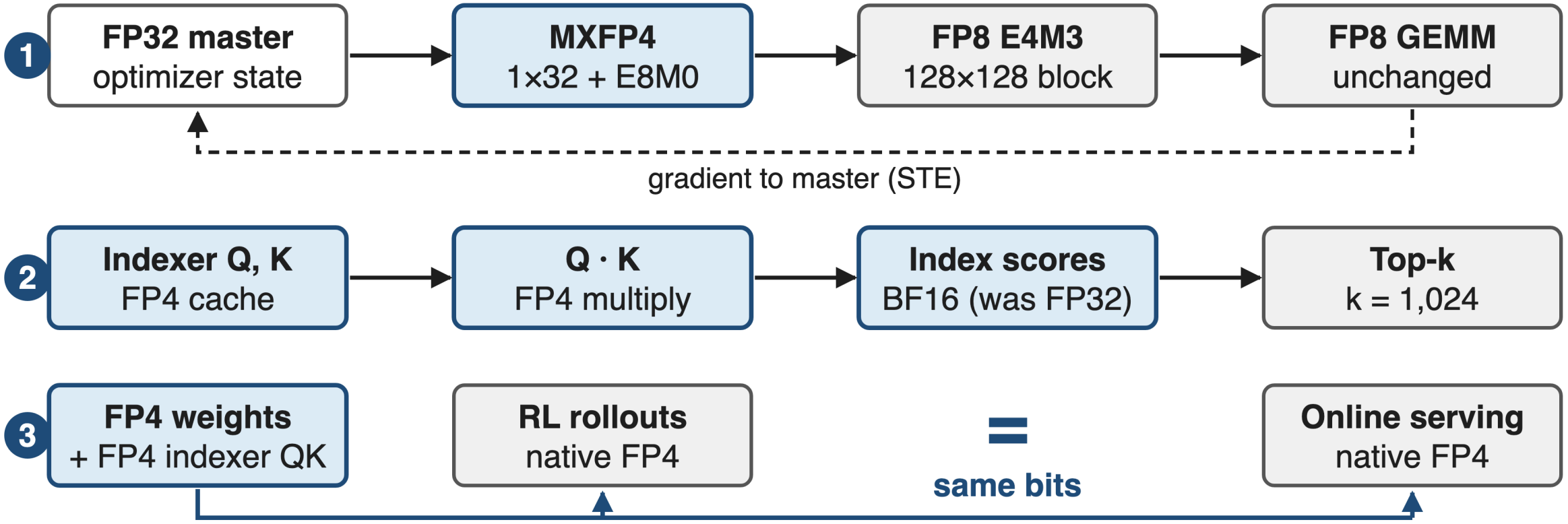
**128 × 128**
FP8 block
one scale

**1 × 32 FP4**
sub-blocks
own E8M0 scale

shade = sub-block scale

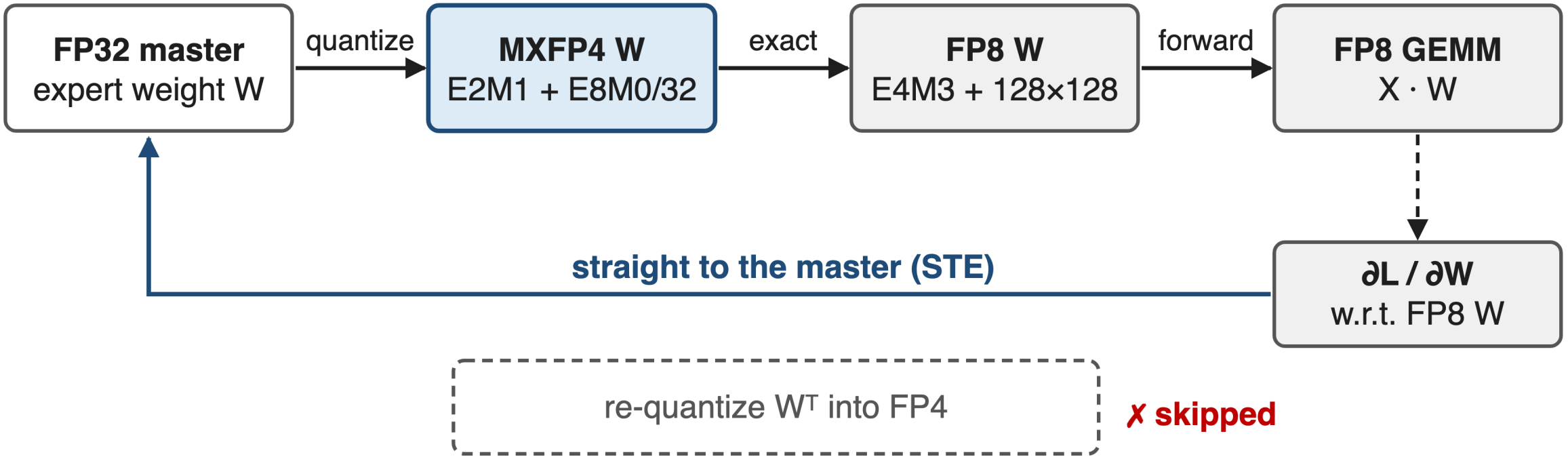
**E4M3 has 2 more exponent bits, so 1×32 FP4 scales fold into a 128×128 FP8 block: QAT reuses the FP8 stack**

# Three FP4 paths, one set of bits



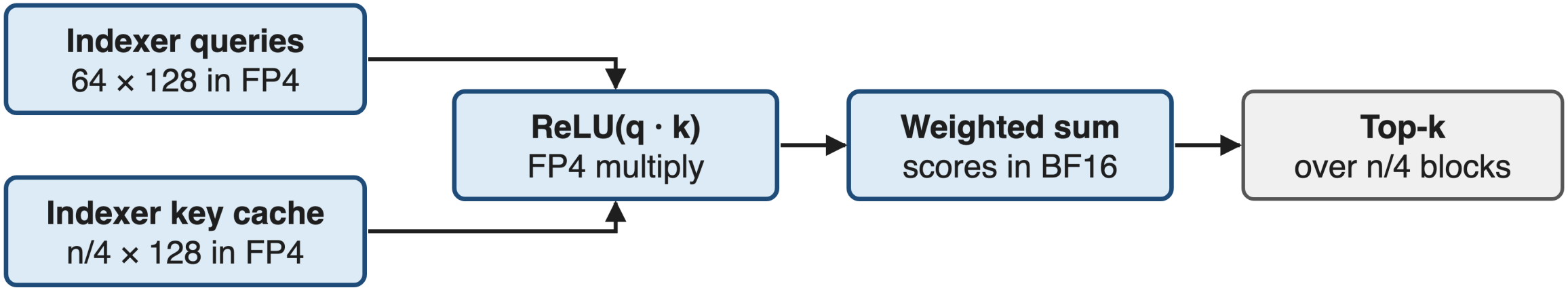

*❶ expert-weight QAT through FP8 · ❷ FP4 indexer QK with BF16 scores · ❸ native FP4 in RL rollouts and serving*

# ❶ Experts train through FP4, compute in FP8




**MXFP4 is 4.25 bits per weight vs 8 in FP8; the backward pass never re-quantizes $W^T$, whose MX scales would sit on the wrong axis**

# ❷ The indexer scores blocks in FP4

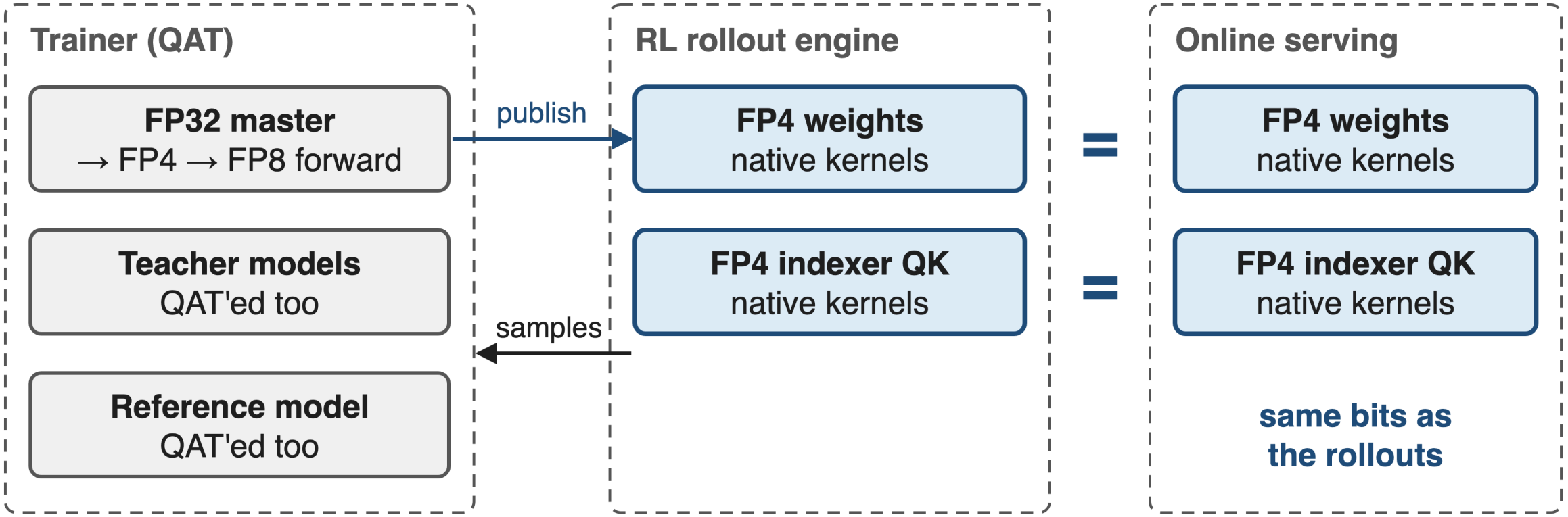



Hadamard rotation before FP4 quantization (reference code)




**At 1M the indexer does 14× a CSA layer's core-attention work (analytic, deck 01): the right place for FP4**

# ❸ Rollouts and serving share the FP4 bits




**RL samples the exact model users get, and native FP4 kernels load about half the expert bytes of FP8**

# Top-k selection runs 2× faster

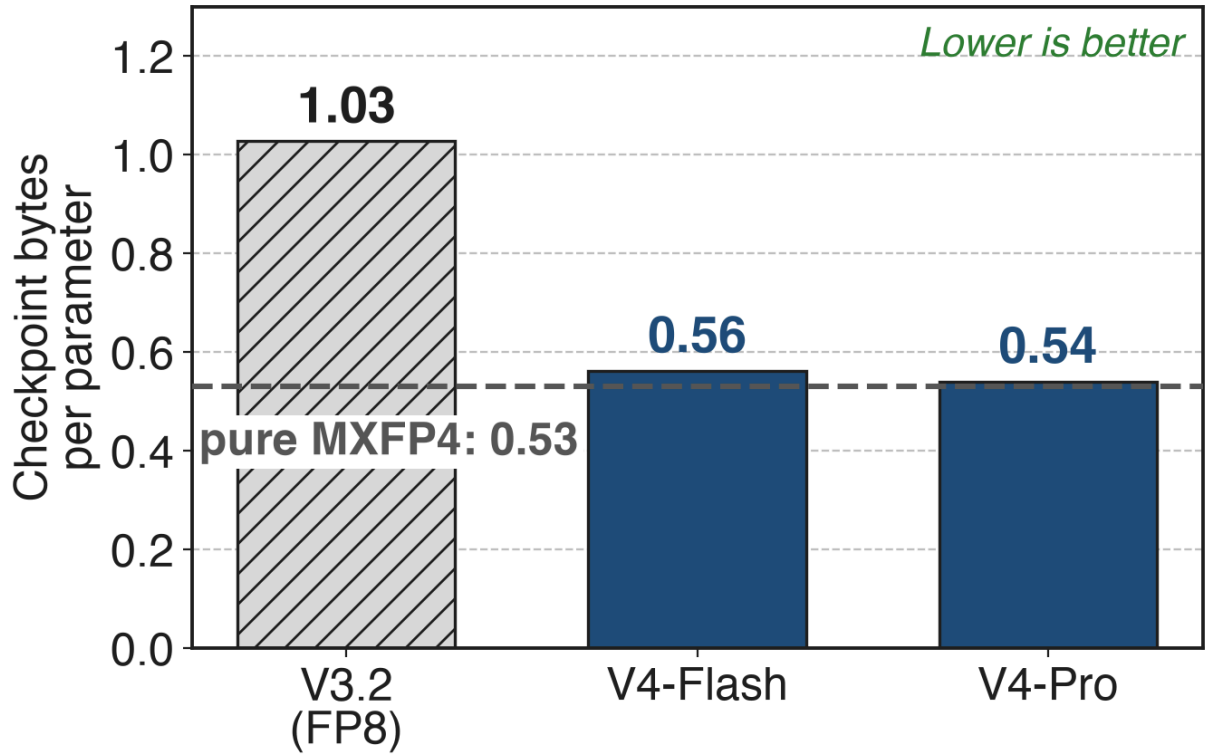
**2×**

faster top-k selector with BF16 index scores, still recalling 99.7% of KV entries

*DeepSeek-V4 §5.2.1; measured by DeepSeek, setup not further specified*

# V4-Pro ships at 0.54 bytes per parameter

- V4-Pro: 1.6T params in 865 GB
- V3.2: 671B params in 690 GB
- Derived: repo size ÷ stated params




**2.4× V3.2's parameters in 1.25× its bytes**

# Cost: no FLOP gain yet, and a scale caveat

- FP4 × FP8 peak = FP8 × FP8 on existing hardware; ~⅓ gain only on future chips
- Exact only while sub-block scales stay close (verified empirically)
- Post-training only; no FP4 vs FP8 accuracy ablation published

**Today's win is memory and bandwidth, not FLOPs**

# Summary

- **Pain**: ~97% of V4-Pro's weights are routed experts (analytic)
- **Insight**: FP4 → FP8 dequantization is exact, so QAT rides the FP8 stack
- ❶ expert QAT + STE ❷ FP4 indexer QK, BF16 scores ❸ native FP4 rollouts = serving
- **2×** faster top-k at **99.7%** recall; **0.54 B** per parameter shipped

https://arxiv.org/abs/2606.19348